Big data in combinatorics: is it feasible?

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For 46 ACC Brisbane 2024

Australia's climate simulator

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Acknowledgements

SageMath, Bliss, Nauty.

Bureau of Meteorology.

Australian National University.

National Computational Infrastructure.

ACCESS-NRI, NCRIS.

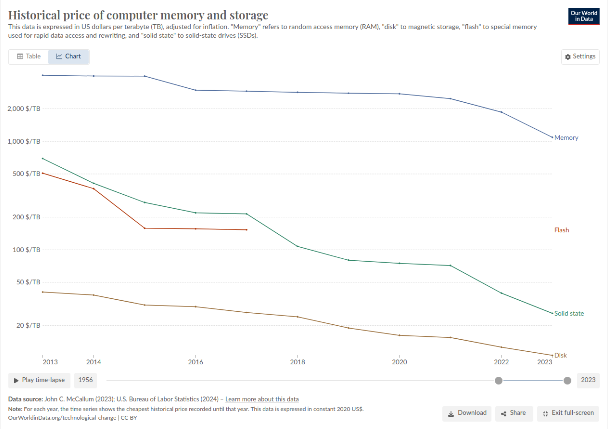
Overview

- How big is big data?
- Some mathematical datasets
- A modest proposal

How big is big data?

Big data is becoming cheaper

Hard disk costs have declined from $41 per Terabyte in 2013 to *$11 per Terabyte* in 2023 in constant US dollars



Historical price of computer memory and storage.
Data source: John C. McCallum (2023); U.S. Bureau of Labor Statistics (2024)
(https://ourworldindata.org/grapher/historical-cost-of-computer-memory-and-storage)

The SKAO: 2 Petabytes per day

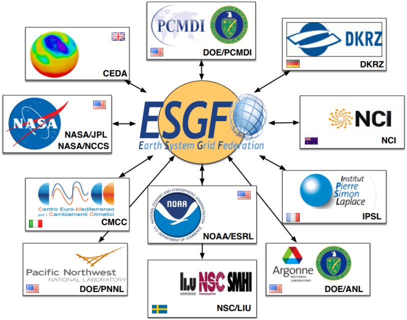
The Square Kilometer Array Observatory will archive about *2 Petabytes* of data *per day*.

CSIRO: Antennas of CSIRO's ASKAP telescope at the Murchison Radio-astronomy Observatory in Western Australia.

(https://www.skao.int/en/explore/big-data 2024)

ESGF: 16+ Petabytes of public databases

The Earth System Grid Federation hosts climate datasets comprising more than *16 Petabytes* of data.



ESGF nodes 2018 (https://e3sm.org/wp-content/uploads/2018/03/ESGF-federation.png)

(Hoffmann et al. 2023)

(https://www.climatemodeling.org/ forrest/presentations/Hoffman_WCRP-OSC-ESGF_20231023.pdf 2023)

CERN ATLAS: 65 Terabyte public database

In 2024 the ATLAS Experiment at CERN released approximately *65 Terabytes* of data collected in 2015 and 2016.

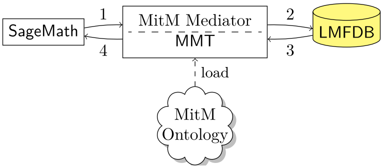
ATLAS open data 2024 (https://atlas.cern/sites/default/files/2024-07/ATLAS-open-data.jpg)

(https://atlas.cern/Updates/News/Open-Data-Research 2024)

Some mathematical datasets

The OpenDreamKit project

The EU funded OpenDreamKit project (2015-2019) included a census of mathematical data sets, and the development of prototypical semantic infrastructure for the FAIR sharing of such data sets. This used the idea of "Deep FAIR": FAIR sharing at the level of mathematical objects.

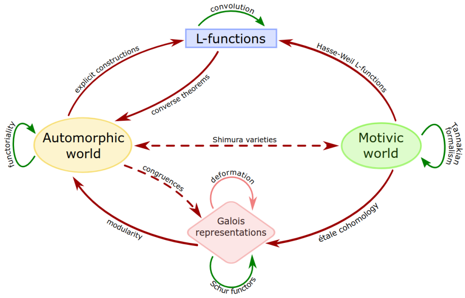


Math in the Middle mediator, OpenDreamKit Project 2019

(OpenDreamKit Project 2015-2019; Berčič, Kohlhase and Rabe 2020)

The L-functions and modular forms database LMFDB

- "The LMFDB is an extensive database of mathematical objects arising in Number Theory."
- Recently extended to include abstract groups.
- About *2 Terabytes* in size.



The LMFDB universe 2024

(The LMFDB Collaboration 2024; Combes et al. 2024)

House of Graphs: a database of interesting graphs

- “It is the aim of this House of Graphs project to find a workable definition of ’interesting’ and provide a searchable database of graphs that conform to this definition.”
- Grew from 1500 to *22 000 graphs* from 2012 to 2023.
- Rebuilt as House of Graphs 2.0 in 2021-2022 to be more flexible and expandable.
- Includes a directory of other graph databases.
- Lean-HoG is a Lean 4 library with an interface to House of Graphs.

(Brinkmann et al. 2013; Coolsaet et al. 2023; Bauer, Berčič and Taslak 2024.)

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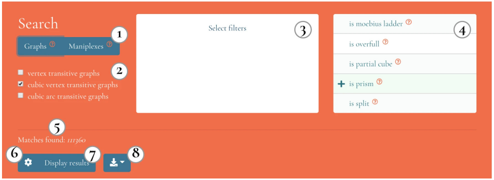
DiscreteZOO: A Fingerprint Database of Discrete Objects

- A data repository, a website and a SageMath Package for discrete combinatorial objects such as graphs with a high degree of symmetry.
- As at 2020 it contains *212 269 graphs*.



DiscreteZoo search box 2020

(Berčič and Vidali 2020)

Datasets of strongly regular graphs

- Spence's lists of strongly regular graphs on at most 64 vertices.
  - Exhaustive lists of non-isomorphic graphs for some tuples of feasible parameters.
  - Flat text files.
- Brouwer's database of parameters of strongly regular graphs.
- Cohen and Pasechnik's Sage implementation of Brouwer's database, with constructions.
  - Example for each tuple of feasible parameters, if known.
  - Sage interface.
- Ihringer's lists of strongly regular graphs on at most 255 vertices.
  - *Over 34 GB* of lists containing more than *200 million graphs* generated by switchings.
  - Gzipped G6 files stored in Dropbox.

(Spence 1995-2006; Brouwer 2008-2017; Cohen and Pasechnik 2016-2017; Ihringer 2022)

Use cases for datasets of strongly regular graphs

1. Test graph recognition algorithms.
   Many papers have been written on testing and extending the Weisfeiler-Lehman algorithm, using data sets of graphs.

2. Obtain statistical distributions of properties of strongly regular graphs.

3. Implement Roberson's classification of strongly regular graphs into four types according to clique number and chromatic number.

(Weisfeiler and Leman 1968; Babai 1980; Yang et al 2024; Cameron 2003; Roberson 2019)

A modest proposal

Bent function Cayley graph data sets (1)

Strongly regular graphs with parameters $(64, 28, 12, 12)$ and $(64, 36, 20, 20)$ as Cayley graphs of bent functions in 6 dimensions. Currently stored **on my desk at home**.

- Database of 11 strongly regular graphs from 4 extended translate (ET) classes.
- SQLite database size is 780 KB.

Bent function Cayley graph data sets (2)

Strongly regular graphs with parameters $(256, 120, 56, 56)$ and $(256, 136, 72, 72)$ as Cayley graphs of bent functions in 8 dimensions. Currently stored **on my desk at home**.

- Bent functions in 8 dimensions up to degree 3.
  - Database of 55 strongly regular graphs from 9 ET classes.
  - SQLite database size is 65 MB.
- Bent functions from the 8 S-boxes of CAST-128.
  - Database of more than *32 million* (32 914 496) strongly regular graphs from 256 ET classes and their duals.
  - SQLite database size is *195 Gigabytes*.
- About *2 billion* strongly regular graphs from the 14 758 ET classes of partial spread bent functions in 8 dimensions and their duals.
  - Currently stored as SageMath sobj files.
  - *Over 6 Terabytes* total size.

Schema for Cayley graph databases

Name	Type	Schema
Tables (4)		
bent_function		CREATE TABLE bent_function(bent_function
bent_function	BLOB	\`bent_function\` BLOB
name	TEXT	\`name\` TEXT UNIQUE
cayley_graph		CREATE TABLE cayley_graph(bent_function
bent_function	BLOB	\`bent_function\` BLOB
cayley_graph_index	INTEGER	\`cayley_graph_index\` INTEGER
graph_id	INTEGER	\`graph_id\` INTEGER
graph		CREATE TABLE graph(graph_id INTEGER, ca
graph_id	INTEGER	\`graph_id\` INTEGER
canonical_label_hash	BLOB	\`canonical_label_hash\` BLOB UNIQUE
canonical_label	TEXT	\`canonical_label\` TEXT
matrices		CREATE TABLE matrices(bent_function BLO
bent_function	BLOB	\`bent_function\` BLOB
b	INTEGER	\`b\` INTEGER
c	INTEGER	\`c\` INTEGER
bent_cayley_graph_index	INTEGER	\`bent_cayley_graph_index\` INTEGER
dual_cayley_graph_index	INTEGER	\`dual_cayley_graph_index\` INTEGER
weight_class	INTEGER	\`weight_class\` INTEGER

Database schema for SQLite version of the Cayley graph database.

Bent function Cayley graph Sage code

CoCalc: Public worksheets, Sage and Python source code

`http://tinyurl.com/Boolean-Cayley-graphs`

GitHub: Sage and Python source code

`https://github.com/penguian/Boolean-Cayley-graphs`

PyPI: Installable Python package

`https://pypi.org/project/boolean-cayley-graphs/`

SourceForge: Documentation of Python modules

`https://boolean-cayley-graphs.sourceforge.io/`

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Possible next steps

- **Find an academic collaborator**.
- Revise and resubmit the paper, arXiv:1705.04507 [math.CO] "Classifying bent functions by their Cayley graphs."
- Arrange for public hosting of the bent function Cayley graph databases.
- Scale up to include the partial spread bent functions.

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NCRIS National Research Infrastructure for Australia

An Australian Government Initiative

Australia's climate simulator